```
 ****************************************************
 *                                                  *
 *           STAAD.Pro                              *
 *           Version  2006    Bld 1002.US           *
 *           Proprietary Program of                 *
 *           Research Engineers,  Intl.             *
 *           Date=    APR  5, 2007                  *
 *           Time=    16:11:22                      *
 *                                                  *
 *      USER ID: Meinhardt Infrastructure Pte Ltd   *
 ****************************************************

     1. STAAD PLANE FRAME 1
INPUT FILE: 12m-EyeBrow.STD
     2. START JOB INFORMATION
     3. JOB NAME - DUBAI METRO PROJECT
     4. JOB CLIENT - GOVERNMENT OF DUBAI _ ROADS & TRANSPORT AUTHORITY
     5. ENGINEER NAME CJA
     6. ENGINEER DATE 06-MAR-06
     7. END JOB INFORMATION
     8. INPUT WIDTH 79
     9. UNIT METER KN
    10. JOINT COORDINATES
    11. 1 0 6.5 0; 2 0.1 6.62 0; 3 0.231 6.7 0; 4 0.93 6.7 0; 5 2 6.7 0; 6 4 6.7
0
    12. 7 6 6.7 0; 8 8 6.7 0; 9 10 6.7 0; 10 12 6.7 0; 11 14 6.7 0; 12 16 6.7 0
    13. 13 18 6.7 0; 14 20 6.7 0; 15 22 6.7 0; 16 23.07 6.7 0; 17 23.769 6.7 0
    14. 18 23.9 6.62 0; 19 24 6.5 0; 101 0 0 0; 201 24 0 0; 202 24 6.015 0
    15. 203 0 6.015 0; 204 0 1.50375 0; 205 0 3.0075 0; 206 0 4.51125 0
    16. 207 24 1.50375 0; 208 24 3.0075 0; 209 24 4.51125 0; 301 12 0 0
    17. MEMBER INCIDENCES
    18. 1 1 2; 2 2 3; 3 3 4; 4 4 5; 5 5 6; 6 6 7; 7 7 8; 8 8 9; 9 9 10; 10 10 11
    19. 11 16 15; 12 17 16; 13 18 17; 14 19 18; 101 101 204; 201 201 207; 202 202
19
    20. 203 203 1; 204 204 205; 205 205 206; 206 206 203; 207 207 208; 208 208
209
    21. 209 209 202; 210 301 10; 211 15 14; 212 14 13; 213 13 12; 214 12 11
    22. MEMBER PROPERTY BRITISH
    23. 1 TO 14 101 201 TO 214 TABLE ST TUB2002006.0
    24. DEFINE MATERIAL START
    25. ISOTROPIC STEEL
    26. E 2.05E+008
    27. POISSON 0.25
    28. DENSITY 77
    29. ALPHA 1.2E-005
    30. DAMP 0.03
    31. END DEFINE MATERIAL
    32. CONSTANTS
    33. *E 2.05E+008 MEMB 1 TO 14 101 201 TO 210
    34. *POISSON 0.25 MEMB 1 TO 14 101 201 TO 210
    35. *DENSITY 77 MEMB 1 TO 14 101 201 TO 210
    36. *ALPHA 1.2E-005 MEMB 1 TO 14 101 201 TO 210
    37. MATERIAL STEEL ALL
    38. SUPPORTS
    39. 101 201 301 PINNED
    40. LOAD 1 SELFWEIGHT
```



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    41. SELFWEIGHT Y -1
    42. LOAD 2 SDL
    43. JOINT LOAD
    44. 1 19 FY -9
```



```
    45. MEMBER LOAD
    46. 1 TO 14 211 TO 214 UNI GY -0.6
    47. LOAD 3 LL
    48. JOINT LOAD
    49. 1 19 FY -3
    50. MEMBER LOAD
    51. 1 TO 14 211 TO 214 UNI GY -0.9
    52. LOAD COMB 1001 1.4DL + 1.6LL
    53. 1 1.4 2 1.4 3 1.6
    54. LOAD COMB 2001 1.0DL + 1.0LL
    55. 1 1.0 2 1.0 3 1.0
    56. LOAD COMB 2002 1.0DL + 0.5LL
    57. 1 1.0 2 1.0 3 0.5
    58. PERFORM ANALYSIS

            P R O B L E M   S T A T I S T I C S
            -----------------------------------

   NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS =     30/    29/     3
   ORIGINAL/FINAL BAND-WIDTH=     22/      2/       9 DOF
   TOTAL PRIMARY LOAD CASES =     3, TOTAL DEGREES OF FREEDOM =     84
   SIZE OF STIFFNESS MATRIX =         1 DOUBLE  KILO-WORDS
   REQRD/AVAIL. DISK SPACE  =      12.1/ 426822.7 MB

    59. LOAD LIST 1001 2001 2002
    60. PARAMETER
    61. CODE BS5950
    62. UNL 2 MEMB 1 TO 14 202 203
    63. UNL 8 MEMB 101 201
    64. LZ 2 MEMB 1 TO 4 11 TO 14
    65. LZ 2 MEMB 5 TO 13
    66. LZ 8 MEMB 101 201
    67. LY 4 MEMB 1 TO 4 11 TO 14
    68. LY 4 MEMB 5 TO 13
    69. LY 10 MEMB 101 201
    70. CHECK CODE MEMB 1 TO 14 101 201 TO 214
STEEL DESIGN

                      STAAD.Pro CODE CHECKING - (BSI )
                      ***********************

 PROGRAM CODE REVISION V2.9_5950-1_2000
```



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ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE          RESULT/   CRITICAL COND/     RATIO/     LOADING/
                            FX            MY             MZ        LOCATION
=======================================================================

     1 ST  TUB2002006.0   PASS      ANNEX I.1          0.207      1001
                         13.53 C          0.00          19.02         -
     2 ST  TUB2002006.0   PASS      ANNEX I.1          0.194      1001
                         10.18 C          0.00          17.87         -
     3 ST  TUB2002006.0   PASS      ANNEX I.1          0.176      1001
                          2.93 C          0.00          16.21         -
     4 ST  TUB2002006.0   PASS      ANNEX I.1          0.075      1001
                          2.93 C          0.00           6.88         -
```

```
  5 ST TUB2002006.0  PASS      ANNEX I.1        0.195      1001
                      2.93 C          0.00      17.99        -
  6 ST TUB2002006.0  PASS      ANNEX I.1        0.224      1001
                      2.93 C          0.00      20.64        -
  7 ST TUB2002006.0  PASS      ANNEX I.1        0.218      1001
                      2.93 C          0.00      20.12        -
  8 ST TUB2002006.0  PASS      ANNEX I.1        0.121      1001
                      2.93 C          0.00      11.13        -
  9 ST TUB2002006.0  PASS      ANNEX I.1        0.436      1001
                      2.93 C          0.00      40.19        -
 10 ST TUB2002006.0  PASS      ANNEX I.1        0.436      1001
                      2.93 C          0.00      40.19        -
 11 ST TUB2002006.0  PASS      ANNEX I.1        0.075      1001
                      2.93 C          0.00       6.88        -
 12 ST TUB2002006.0  PASS      ANNEX I.1        0.176      1001
                      2.93 C          0.00      16.21        -
 13 ST TUB2002006.0  PASS      ANNEX I.1        0.194      1001
                     10.18 C          0.00      17.87        -
 14 ST TUB2002006.0  PASS      ANNEX I.1        0.207      1001
                     13.53 C          0.00      19.02        -
101 ST TUB2002006.0  PASS      BS-4.7   (C)     0.072      1001
                     35.81 C          0.00       0.00         0.00
201 ST TUB2002006.0  PASS      BS-4.7   (C)     0.072      1001
                     35.81 C          0.00       0.00         0.00
202 ST TUB2002006.0  PASS      ANNEX I.1        0.207      1001
                     32.81 C          0.00      19.02        -
203 ST TUB2002006.0  PASS      ANNEX I.1        0.207      1001
                     32.81 C          0.00      19.02        -
204 ST TUB2002006.0  PASS      ANNEX I.1        0.096      1001
                     35.06 C          0.00       8.80        -
205 ST TUB2002006.0  PASS      ANNEX I.1        0.144      1001
                     34.31 C          0.00      13.20        -
206 ST TUB2002006.0  PASS      ANNEX I.1        0.191      1001
                     33.56 C          0.00      17.60        -
207 ST TUB2002006.0  PASS      ANNEX I.1        0.096      1001
                     35.06 C          0.00       8.80        -
208 ST TUB2002006.0  PASS      ANNEX I.1        0.144      1001
                     34.31 C          0.00      13.20        -
209 ST TUB2002006.0  PASS      ANNEX I.1        0.191      1001
                     33.56 C          0.00      17.60        -
```



```
ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE          RESULT/   CRITICAL COND/     RATIO/     LOADING/
                            FX            MY             MZ       LOCATION
=======================================================================

210 ST TUB2002006.0  PASS      BS-4.7   (C)     0.045      1001
                     40.11 C          0.00       0.00         0.00
211 ST TUB2002006.0  PASS      ANNEX I.1        0.195      1001
                      2.93 C          0.00      17.99        -
212 ST TUB2002006.0  PASS      ANNEX I.1        0.224      1001
                      2.93 C          0.00      20.64        -
213 ST TUB2002006.0  PASS      ANNEX I.1        0.218      1001
                      2.93 C          0.00      20.12        -
214 ST TUB2002006.0  PASS      ANNEX I.1        0.121      1001
                      2.93 C          0.00      11.13        -

  ************** END OF TABULATED RESULT OF DESIGN **************

   71. LOAD LIST ALL
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   72. PRINT SUPPORT REACTION LIST 101 201 301
 SUPPORT  REACTION LIST      101
```



```
  SUPPORT REACTIONS -UNIT KN    METE     STRUCTURE TYPE = PLANE
  -----------------

 JOINT  LOAD    FORCE-X    FORCE-Y    FORCE-Z      MOM-X      MOM-Y      MOM Z

   101     1      0.38       4.26       0.00       0.00       0.00       0.00
           2      0.63      12.28       0.00       0.00       0.00       0.00
           3      0.95       7.91       0.00       0.00       0.00       0.00
        1001      2.93      35.81       0.00       0.00       0.00       0.00
        2001      1.95      24.45       0.00       0.00       0.00       0.00
        2002      1.48      20.49       0.00       0.00       0.00       0.00
   201     1     -0.38       4.26       0.00       0.00       0.00       0.00
           2     -0.63      12.28       0.00       0.00       0.00       0.00
           3     -0.95       7.91       0.00       0.00       0.00       0.00
        1001     -2.93      35.81       0.00       0.00       0.00       0.00
        2001     -1.95      24.45       0.00       0.00       0.00       0.00
        2002     -1.48      20.49       0.00       0.00       0.00       0.00
   301     1      0.00       7.09       0.00       0.00       0.00       0.00
           2      0.00       7.94       0.00       0.00       0.00       0.00
           3      0.00      11.91       0.00       0.00       0.00       0.00
        1001      0.00      40.11       0.00       0.00       0.00       0.00
        2001      0.00      26.95       0.00       0.00       0.00       0.00
        2002      0.00      20.99       0.00       0.00       0.00       0.00

  ************** END OF LATEST ANALYSIS RESULT **************

   73. LOAD LIST ALL
   74. PRINT FORCE ENVELOPE NSECTION 5 LIST 1 TO 14 101 201 TO 214
 FORCE    ENVELOPE NSECTION 5
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     MEMBER FORCE ENVELOPE
     ---------------------

     ALL UNITS ARE KN   METE

  MEMB  DISTANCE          FY    LD         MZ    LD       FZ    LD        MY    LD

    1     0.00  MAX     7.46 1001      19.02 1001     0.00 2002      0.00 2002
                MIN     0.96    1       2.44    1     0.00 2002      0.00 2002
          0.03  MAX     7.41 1001      18.79 1001     0.00 2002      0.00 2002
                MIN     0.95    1       2.41    1     0.00 2002      0.00 2002
          0.06  MAX     7.35 1001      18.56 1001     0.00 2002      0.00 2002
                MIN     0.94    1       2.38    1     0.00 2002      0.00 2002
          0.09  MAX     7.30 1001      18.33 1001     0.00 2002      0.00 2002
                MIN     0.93    1       2.35    1     0.00 2002      0.00 2002
          0.12  MAX     7.24 1001      18.10 1001     0.00 2002      0.00 2002
                MIN     0.93    1       2.32    1     0.00 2002      0.00 2002
          0.16  MAX     7.19 1001      17.87 1001     0.00 2002      0.00 2002
                MIN     0.92    1       2.29    1     0.00 2002      0.00 2002

 |---------------------------------------------------------------------------
 | MAX/MIN FORCE VALUES FOR MEMB      1, AMONGST ALL SECT LOCATIONS
```